

Utsav Munendra

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EDUCATION

University of California, Los Angeles

Sept 2018 - Dec 2021

B.S. in Computer Science, Henry Samueli School of Engineering and Applied Science

GPA: 3.96/4.00, *summa cum laude*

Tutoring Chair at UCLA Upsilon Pi Epsilon (UPE) CS Honor Society

Student Director at Grad2Mentor

TEACHING

Reader and Undergraduate Teaching Assistant

[University of California, Los Angeles](#)

CS33: Computer Organization (Systems Architecture)

Completed LA Pedagogy Seminar on inclusive STEM teaching by facilitating collaborative and inclusive learning.

PROJECTS

WebAssembly Runtime (WAMR) for ARM Mbed

2020

Allows existing WebAssembly modules to run on ARM Mbed boards and the web alike for increased portability for IoT applications.

UCLA Cuneiform Digital Library Initiative Research Lab

2019

The Cuneiform Digital Library Initiative (CDLI) maintains a database of over 334,000 Assyriological artifacts recording each artifact's period, place of origin and tablet writings. Built a user-customizable [heatmap](#) combining temporal, geographic and textual data to render informative visualizations assisting Assyriologists in their research.

AWARDS

Dean's Honor List, 2018 - 2021

[University of California, Los Angeles](#)

Secured A grade for the quarter with extra course load. Awarded for 7 quarters.

AP Scholar with Distinction, 2018

The College Board

Achieved a score of 5/5 in six AP exams.

SOF Cyber Olympiad, 2014 - 2017

[Science Olympiad Foundation](#)

International Rank 1 (Gold Medal): 14th (2014), 15th (2015), 17th (2017) SOF Cyber Olympiad

International Rank 3 (Bronze Medal): 16th (2016) SOF Cyber Olympiad

IIT-KGP Kshitij Teenage Prodigy Contest, 2018

Indian Institute of Technology, Kharagpur

Ranked second for essay analyzing the progress and future of artificial intelligence, selected from hundreds of submissions.

IIT-K Techkriti Open School Championship, 2016

Indian Institute of Technology, Kanpur

Ranked second for paper on "[Clean Ganga: Harnessing Natural Resources to Rejuvenate the River](#)." Used mathematical models to delineate a new course of action for cleaning the Ganga River. Surveyed 200+ students to highlight discrepancies in social perceptions of river cleaning. Defended the paper before geology professors and PhD students of IIT-K.

WORK EXPERIENCE

Meta, Senior Software Engineer

Jan 2022 - Present

Virtual Reality Firmware

Burlingame, CA

- Brought up boards, sensors and communication interfaces for [Meta Quest 3/3S](#) headsets and [Meta Quest Touch Plus](#) controllers.
- Brought up USB Mass Storage end-to-end solution, from AOSP filesystem integration to flash sensor drivers. Committed bugfixes to LittleFS and Zephyr.
- Designed sensor driver framework and HAL implementations for efficient sensor sampling from Qualcomm Hexagon DSP while targeting reusability with other MCUs.
- Improved compute load and modularity of sensor data streaming APIs via Linux kernel driver optimizations.
- Led in-market support for camera firmware involving architecture improvements, agentic bug-resolution, AI-assisted data-analysis along with collaboration with Qualcomm on CamX.
- Led JMD as the sole partner SWE for [Meta Quest Controller Rechargeable Batteries](#), and designed its interface and protocol with OEM controllers.
- Assisted various factory builds via on-site debug and bringup support along with tooling implementation.

Green Hills Software, Software Engineering Intern

Summer 2021, Summer 2022

MULTI Debugger & Advanced Products Group

Santa Barbara, CA

- Extended the Rust compiler and standard library to natively support new Green Hills' Integrity RTOS targets. Proved support by hosting a web server on targets backed by prevalent async runtime Rust crates.
- Increased performance by 4% of GHS MULTI Debugger by making IPC more efficient.
- Debugged a SoC failing GHS nightly validations.
- Built simplified software models of GHS Trace Decoders to characterize the performance difference seen when decoders were run on Windows vs. Linux.

Verkada, Software Engineering Intern

Winter 2021, Spring 2021

Cameras Core OS

San Mateo, CA

- Wrote Go services to mount filesystems, partition storage and set up encrypted volumes.
- Reproduced a bootloop bug on low power, and assisted hardware team by making tools to log SD card temperature.
- Assisted in board bringup for [CD62](#) Dome camera by modifying BitBake recipes and created test racks for new camera models.
- Fixed bugs across Python, Go and C++ codebase.

OPEN SOURCE

Zephyr RTOS

2022 - 2025

Approved contributor for Zephyr Project, merged patches for various flash, MSPI driver and compatibility layer fixes. Committed bugfixes to LittleFS upstream as part of Meta's firmware work.

SKILLS

Languages C/C++, Python, Rust, Go

RTOS Zephyr, RTX5, Qualcomm QuRT

Tools GDB, CMake, Makefile, BitBake, Bazel, SCons, Logic Analyzer, Oscilloscope